1. (i) $\varphi(7)=6$. $20 \equiv 2 \pmod 6$. $5^{20} \equiv 5^2 \equiv 4 \pmod 7$

(ii) $2^{16} \equiv 0 \pmod 8$

(iii) $\varphi(11)=10$. $1001 \equiv 1 \pmod{10}$. $7^{1001} \equiv 7 \pmod{11}$

(iv) $6^{10} \equiv 1 \pmod 5$; $6^{10} \equiv 0 \pmod 3$ $\Rightarrow$ $6^{10} \equiv 6 \pmod{15}$

2. If the current number is $\overset{(\times 3)}{14}$, $\overset{(\times 2)}{21}$ or $\overset{(\times 5)}{33}$, then the player to move next is winning.

A "dying" number is such that no matter what this player do, he will always obtain 14, 21 or 33 (and then lose the game).

Claim: "dying" number does not exists.

Claim: If a player has a winning strategy, then he must leave a dying number to his opponent at sometime.

Conclusion: no one has a winning strategy.

3. Let $n=pm$. Then $\gcd(m,p)=1$. Furthermore, $\varphi(n) = (p-1)\cdot\varphi(m)$.

[If $\gcd(k,l)=1$, then $\varphi(kl)=\varphi(k)\varphi(l)$.]

Hence, $p^{\varphi(n)} = (p^{\varphi(m)})^{p-1} \equiv 1 \pmod m$, or $p^{\varphi(n)} = km+1$.

Thus, $p^{\varphi(n)+1} = kmp+p = kn+p \equiv p \pmod n$.

4. $n=87$. $\varphi(n)=56$. $d=19 \Rightarrow ed \equiv 1 \pmod{\varphi(n)}$ gives $e=3$.

Thus, $4^3 \equiv 64 \pmod{87} \Rightarrow$ fo
$13^3 \equiv 43 \pmod{87} \Rightarrow$ od

$\Rightarrow$ food.